

# Qatar-1b

R-2724

Benchmark run · workflow W8-benchmark-deep · record deep-bench\_Qatar-1\_160904 · created 2026-09-01T23:23:54+00:00

## BENCHMARK: NON-DETECTION

### Results

|                                                   |                                   |
|---------------------------------------------------|-----------------------------------|
| Mid-Transit Time (Tmid)                           | 2457635.66629 +/- 0.00088 BJD_TDB |
| Ratio of Planet to Stellar Radius (Rp/R*)         | 0.192 +/- 0.056                   |
| Transit depth (Rp/Rs)^2                           | 3.69 +/- 2.17 [%]                 |
| Orbital Inclination (inc)                         | 88.64 +/- 2.85                    |
| Airmass coefficient 1 (a1)                        | 34.95 +/- 16.49                   |
| Airmass coefficient 2 (a2)                        | -2.92 +/- 1.42                    |
| Scatter in the residuals of the lightcurve fit is | 47.86 %                           |
| Best Comparison Star                              | #2 - [459, 284]                   |
| Optimal Aperture                                  | 3.0                               |
| Optimal Annulus                                   | 7.5                               |
| Transit Duration (day)                            | 0.0825 +/- 0.0062                 |

### Accuracy vs published ephemeris (ExoClock/NEA)

|                                      |                                                        |
|--------------------------------------|--------------------------------------------------------|
| Rp/R $\blacksquare$ fit vs published | 0.192 $\pm$ 0.056 vs 0.1463 (deviation 0.63 $\sigma$ ) |
| Duration fit vs published            | 0.0825 d vs 0.0692 d (deviation 1.66 $\sigma$ )        |
| Mid-time O-C                         | -1.4 min                                               |
| Depth SNR                            | 2.6                                                    |
| Residual scatter                     | 47.86 %                                                |

### Light curve

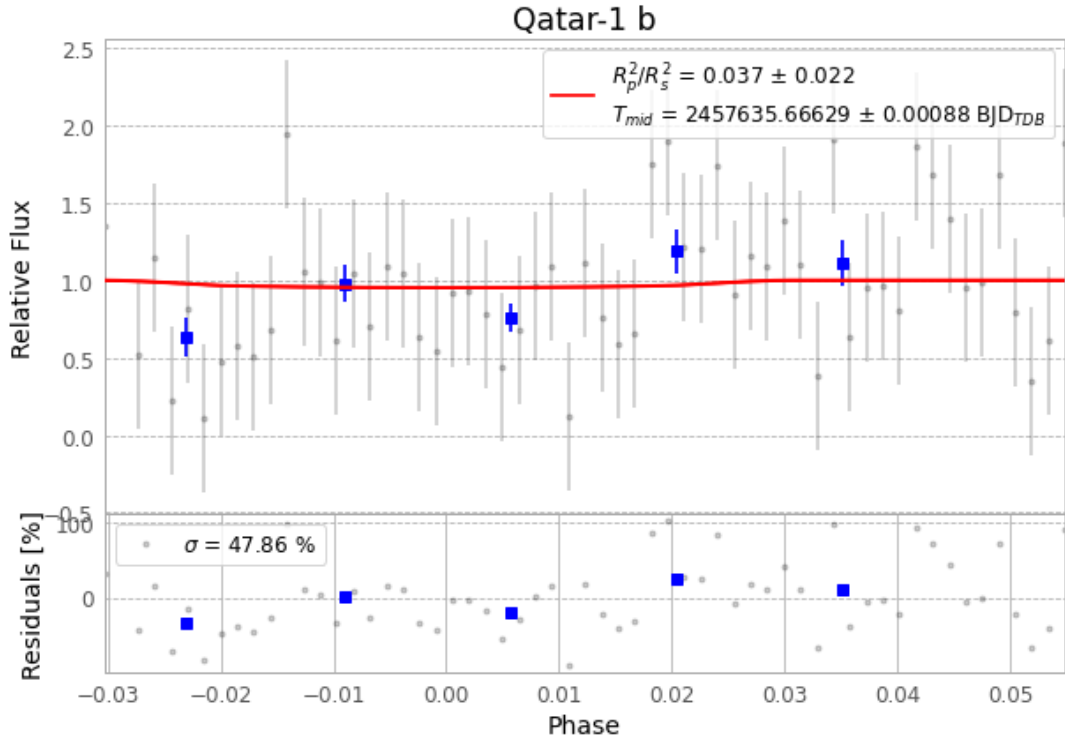


Figure 1 — FinalLightCurve\_Qatar-1 b\_2016-09-03.png (SHA-256 dd4f6477678ab917...)

## Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [170, 327], 3 comparison star(s). Exact configuration: parameters.inits\_json in record.json (SHA-256 dadae0ad8c7a2428...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline\_commit a4253ce

## Data provenance

58 FITS frames (38 MB), Qatar-1160904025411.FITS → Qatar-1160904054812.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-160904.log (cd64aca9af2d...), FinalLightCurve\_Qatar-1 b\_2016-09-03.png (dd4f6477678a...), FinalLightCurve\_Qatar-1 b\_2016-09-03.csv (68d88e7d2c2c...)

## Compute resources

Wall time 151.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

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Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-09-01 23:23Z.